

Math 1 • Repetition re-replace review

Only B19, C21, D13, D22 after the second replacement pass. For checking — not a full paper.

B19 • REPLACE AGAIN

was repeating B3 dice → quadratic real-roots template

The line L_1 passes through the point $A(1, 2)$ and is perpendicular to the line segment connecting A and $B(5, 4)$. The line L_2 passes through B and is perpendicular to the line $x + 3y = 0$. The lines L_1 and L_2 intersect at the point C .

What is the area of the triangle ABC ?

- A 5
- B 10
- C 20
- D 30
- E 40

C21 • REPLACE AGAIN

tank/pipe/flow/rate overrepresented across A–E

Find the set of values of k for which the equation

$$k = \frac{2x^2 + 3}{x^2 - 1}$$

has real solutions for x .

- A $k \leq -3$ or $k > 2$
- B $-3 \leq k < 2$
- C $k \leq -3$ or $k \geq 2$
- D $k < -3$ or $k > 2$
- E $k \geq -3$ or $k < 2$
- F $-3 \leq k \leq 2$

D13 • REPLACE AGAIN

another three-dice probability; Mock D already probability-heavy

Let P be a quadrilateral that has rotational symmetry of order greater than 1, but no line symmetry.

Let Q be a quadrilateral that has one or more lines of symmetry, but has rotational symmetry of order 1.

Which of the following is a possible pair (P, Q)?

(Assume any named shape is a non-special case, e.g. a 'rectangle' is not a square.)

- A** (Parallelogram, Kite)
- B** (Kite, Parallelogram)
- C** (Rectangle, Kite)
- D** (Parallelogram, Rhombus)
- E** (Rhombus, Rectangle)
- F** (Square, Isosceles Trapezium)

D22 • REPLACE AGAIN

density overrepresented in Mock D

Which of the following describes the set of real values of x that satisfy the inequality?

$$\frac{(x-1)^2(2-x)}{(x+3)(x^2+1)} \leq 0$$

- A** $-3 < x \leq 2$
- B** $x < -3$ or $x \geq 2$
- C** $x < -3$ or $1 \leq x \leq 2$
- D** $x < -3$ or $x = 1$ or $x \geq 2$
- E** $x \leq -3$ or $x = 1$ or $x \geq 2$